

Chaotic Circuit

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(Dated: 17:07 Monday 17th November, 2025)

In this report I will explore the behaviour of the chaotic circuit. Simple chaotic behaviour will be exhibited in basic plots such as phase space and Poincare section plots through the lens of a driven pendulum. The regimes of chaos and nonchaos will be clearly defined for chaotic systems through the behaviour of the pendulum in the plots. The circuit proposed by Kiers et al. will be the main subject of chaotic investigation. A similar circuit is built and then tested for chaotic behaviour. More advanced techniques in chaotic analysis will be used such as bifurcation plots and Lyapunov exponents to better define chaotic regimes. Lastly, through computation of Lyapunov exponents, the dimensionality of the circuit will be calculated and compared to the value put forth by Kiers.

INTRODUCTION

"Given for one instant an intelligence which could comprehend all the forces by which nature is animated and the respective situation of the beings who compose it-an intelligence sufficiently vast to submit these data to analysis-it would embrace in the same formula the movements of the greatest bodies of the universe and those of the lightest atom; for it, nothing would be uncertain and the future, as the past, would be present to its eyes."

- Pierre Simon Laplace, *A philosophical essay on probabilities*, pg. 4 [1]

The topic of determinism has been discussed by philosophers and physicists alike for millenia and will continue for the next thousands of years [2]. In a philosophical sense, determinism is more about free will and moral responsibility of human beings for their actions, whereas the interpretation in physical sciences, relates to predictions given the present state of systems. It is outside the scope of this report to, firstly, argue the necessity for determinism in governing this universe and, secondly, discuss the interplay of free will and determinism, we will leave this to contemporary philosophers and budding online intellectuals.

In the realm of chaos theory, I will assume that determinism is the correct interpretation of our universe, that is, to assume that the demon is real and computationally possible. The definition of determinism I will be abiding by is one given by Hoefer's paper on "Causal Determinism" (2003), [3], where determinism is loosely encompassed by the definition:

Determinism is true of the world if and only if, given a specified way things are at a time t , the way things go thereafter is fixed as a matter of natural law.

In other words, the world this report resides in, is one governed solely by Newton's classical laws of physics. I will ignore the challenges to determinism that quantum mechanics brings [4] and instead focus on the certainty of states given their initial conditions. The study of chaos is, counter-intuitively, not probabilistic. Chaotic systems do not have different outcomes if a perfect process is repeated. Instead, the

different outcomes only appear to us humans because of our imperfect knowledge of the parameters of the initial state of the system. Therefore, to an all knowing demon who is able to perceive the parameters of a system to infinite figures, the future and the past, are one and the same.

The language of chaos is often, understandably so, misused as a synonym for randomness. This comes from their shared characteristic of unpredictability however the root cause of this unpredictability is not the same. Chaotic systems are inherently unpredictable due to our lack of mathematical precision and the system's incredible sensitivity to initial conditions. The time evolution of these systems may or may not exhibit patterns to us which also propagates the notion of randomness however they are not equivalent. It is important to clear this language as throughout this report, I may describe trajectories or shapes as disordered and lacking form but this does not mean it is not governed by deterministic physical law.

The conditions for chaos are well understood, being formulated at the time of the discovery of chaotic behaviour by Edward Lorenz in 1963 [5]. The discovery was serendipitous and as the story goes, Lorenz was trying to model weather patterns using the state space of a 12 component model. Each component possessed a differential equation relating to a real world atmospheric parameter such as temperature, pressure etc. He tried to solve this model numerically however found that his model was nonperiodic whereas most studied solutions were periodic.

The famous story in his career is that one day, he wanted to examine one of the solutions in greater detail. He had inputted in 12 numbers that the simulation had printed out, effectively restarting the process from about half way and found that his two runs had completely different outputs despite their shared initial conditions. Later, he would attribute the truncated output to be the source of this divergence [6].

The computer he used for the simulation stored the output numbers to six decimal places however when he printed them out to restart the process, he only printed up to three decimal places to save paper. This one part in ten thousand difference had caused the simulation to wildly alter its course from its original trajectory [7]. The first condition of chaotic motion is

then the system must exhibit sensitive dependence on its initial conditions.

The story then goes that while visiting his colleague, Barry Saltzman, he found that in Barry's work, there was a three component subset that displayed aperiodicity, or chaotic motion. This three component model became the famous Lorenz model for which chaotic motion can arise [8]. The beauty of the Lorenz equations came from their simplicity, only requiring three variables for the properties of chaos to appear. Through his work and shown in his paper in 1963, another condition for the aperiodicity is the equations must be non-linear.

Remarkably, the development of chaos theory since its discovery in 1963, continues on the work of Newton in his final pursuit for a solution in the three body problem. Coincidentally, the three body problem has conditions that align with the ones set out by chaos theory. Newton successfully solved the two body problem when he introduced to the world his idea of gravitation. In that scenario, he only needed to worry about the motion of two celestial objects and therefore, the problem could be solved analytically and the solution exhibited periodicity and stability always. However, the inclusion of a third gravitational body would introduce a non-linear term into the equations as well as increasing the variable count to three, satisfying all conditions for aperiodicity and therefore the solution to Newton's three body problem is chaotic [9].

The word 'chaos' was first coined by Jim Yorke to describe types of non-linear dynamical systems however the term has transcended the technical community. The idea of chaos and its child, 'The Butterfly Effect', has become apart of everyday language and integrated itself seamlessly in pop culture. It is common to hear in economics and political discussions, the term 'bifurcation' in relation to causal relationships, especially one where predictability of outcomes is at its lowest. There is a Wikipedia article dedicated to the Butterfly Effect in popular culture as the metaphor of a butterfly flapping its wings to cause a tornado in Texas (presented by Lorenz in a 1972 talk) resonates with the general public and their perception of the world around them. The imagery of the smallest detail influencing the world's most significant events is one of the most powerful narratives to ever been told. But the best kind of stories are the ones that did actually happen and its our marvel of the possibility turning into reality is the reason why the butterfly effect and chaos theory really has a special place in our society today.

THEORY

The central characteristic of chaos is that the system does not repeat its past behaviour, not even approximately. Despite their lack of regularity, the aperiodicity still follow the deterministic equations derived from Newton's laws.

Conditions for chaos

As mentioned before, Edward Lorenz had come up with the two conditions for chaotic motion. For a system to be chaotic, it must:

1. be described by at least three components that are independent dynamical variables
2. contain a non-linear term in its equations of motions that couples several of the variables

The sensitivity to initial conditions is a consequence of the second condition. The non-linear term grows errors in predictions exponentially in time so any small shifts from its original state will cause that state to become essentially unrecognisable in evolution. It is this consequence that will drive the motivation for investigation in chaos theory. Another non-trivial consequence of the second condition is that differential equations containing non-linear terms are impossible to solve analytically and so solutions for equations of motion will have to be done numerically.

Phase space

To study dynamical systems, often phase spaces are used to describe states and their evolution. The phase space is a mathematical space with orthogonal coordinate directions representing each of the variables required to define the state of the system. Commonly, the axis of the phase space will come in position, velocity pairs or analogues. For each solution on a phase space diagram, there will be a corresponding phase trajectory.

There is an extremely non-trivial and significant property of these trajectories, being they are non-crossing. Two trajectories, no matter how close, will never cross each other. This is a consequence of determinism, once a state of the system is known, its time evolution into the future and in the past are known as well. A state in two trajectories would render the system indeterminate and therefore contradict the uniqueness of the determinism on which the motion is formulated upon.

Phase spaces of conservative (constant energy) systems preserve the areas in the space. All the points in a given area of phase space will move in such a way to preserve the area occupied by these same points at some later time. If the system does not conserve energy, the phase area or volume will contract. This is known as a dissipative dynamical system and they introduce a mathematical object known as an attractor. The point is defined as the end state, or fully time evolved state, for all finite states that describes the system.

In order to simplify complicated phase space diagrams, Poincare sections are used. They are constructed by viewing the phase space diagram stroboscopically in such a way that the motion is observed

periodically. Therefore, the Poincare section will provide information about the ratio of the strobing and natural frequency of the dynamical motion. If the ratio is rational such that the natural frequency ω_0 is equal to $\frac{p}{q}\omega_s$ where ω_s is the strobing frequency and p and q are integers, then there are q points on the diagram. If the ratio is irrational, then no points are repeated on the plot and so a filled in line of the trajectory will appear.

Lyapunov Exponents

The sensitivity to initial conditions can be measured using a Lyapunov exponent. The exponent, λ , can be calculated using the divergence of two slightly differing initial states, x and $x + \epsilon$. After n iterations, their divergence is characterised approximately as [10]

$$\epsilon(n) \approx \epsilon e^{\lambda n}. \quad (1)$$

The error growth is, aforementioned, characterised as an exponential and the Lyapunov exponent, λ , describes the average rate of divergence. The average needs to be calculated over many initial conditions spread over the trajectory. If λ is negative then the slightly separated trajectories will converge and the evolution is not chaotic but if λ is positive then the evolution is sensitive to initial conditions and is chaotic. To find λ , we consider the one dimensional map given by $x_{n+1} = f(x_n)$. The difference between two initially nearby states after the n th step is

$$f^n(x + \epsilon) - f^n(x) \approx \epsilon e^{n\lambda} \quad (2)$$

We can rearrange for λ ,

$$\lambda \approx \frac{1}{n} \log_e \left[\frac{f^n(x + \epsilon) - f^n(x)}{\epsilon} \right].$$

For small ϵ , the expression inside the brackets reduces down to the derivative of f^n with respect to x , from first principles,

$$\lambda \approx \frac{1}{n} \log_e \left| \frac{df^n}{dx} \right|. \quad (3)$$

Here, we take the derivative through using the chain rule for the n th iterate and take the limit as n goes to infinity,

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log_e |f'(x_i)|$$

Dimensionality

So far I have only considered calculating Lyapunov exponents in one dimension. For higher-dimensional

systems, the process of obtaining these exponents is non-trivial. It may be helpful to revisit the idea and motivation behind these exponents. The Lyapunov exponents measure the average rate of divergence between neighbouring trajectories on the attractor. The divergence is measured as the distance between the trajectories and so can only be expressed for an one dimensional comparison. If we want to measure the divergence of a two dimensional trajectory, then we require two Lyapunov exponents, one for each dimension. The exponents then describe the rate of divergence in uniquely defined, orthogonal directions. Therefore, for n -dimensions, we require n Lyapunov exponents.

We can imagine the a small n -dimensional sphere centered at some point on the attractor and the surface of which consists of phase points from nearby trajectories. The sphere will evolve into an ellipsoid in time with different axes contracting and expanding. The average rates of expansion or contraction are the Lyapunov exponents and for the i th axis, the corresponding exponent is defined as

$$\lambda_i = \lim_{t \rightarrow \infty} \log_e \left[\frac{L_i(t)}{L_i(0)} \right] \quad (4)$$

where $L_i(t)$ is the radius of the ellipsoid along the i th axis. If we consider, in $\mathbb{R}^3$, a small volume in phase space. The volume will change in time as

$$V(t) = V_0 e^{(\lambda_1 + \lambda_2 + \lambda_3)t}. \quad (5)$$

Therefore, the rate of change of volume is found by

$$\frac{dV}{dt} = \sum_{i=1}^3 \lambda_i V(t). \quad (6)$$

We can combine it with the divergence theorem, or Gauss's theorem to obtain a relationship for the sum of the exponents, that is,

$$\frac{dV}{dt} = \int \nabla \cdot \mathbf{F} dV = \sum_{i=1}^3 \lambda_i V(t). \quad (7)$$

I need to talk a step back here and talk about how to actually obtain λ_i because not only would neighbouring points on nearby trajectories involve points along the same dimensional axis, we need to consider points off-axis as well and so our one dimensional view of Lyapunov exponents will no longer function here. I first have to define what I mean by dimension. In this section, I will be talking about *capacity dimension*. Consider a d dimensional line, square, cube analogue with side length, L . We can split this object into N pieces with side length, ϵ . For the n th split, there will be $(2^d)^n$ objects with side length $\frac{L}{2^n}$. We can trivially obtain the number of objects N as equal to the ratio of L and ϵ to the d th power, that is $N(\epsilon) = \left(\frac{L}{\epsilon}\right)^d$. If

we rearrange for d and take ϵ to go to zero, or equivalently, $n \rightarrow \infty$, we obtain the definition for capacity dimension,

$$d_C = \lim_{\epsilon \rightarrow 0} \frac{\log_e N(\epsilon)}{\log_e \frac{1}{\epsilon}}. \quad (8)$$

The biggest assumption here is that we can take infinite iterations of the set however in practice we are limited, computationally, to a finite set that is a subset of the infinite set (an implicit assumption). Experimentally, we can estimate the capacity dimension by using a technique called box counting where a map is covered with boxes of side length ϵ which is varied to obtain multiple values for $N(\epsilon)$. Then the graph of $\log_e N(\epsilon)$ vs $\log_e \frac{1}{\epsilon}$ can be plotted as $\epsilon \rightarrow 0$ and the gradient estimated to be d_C .

However, as you may observe from the methodology, this method of defining dimensionality does not account for the changing density in the regions of the boxes, only counting the number of points inside the box. There are two other dimension quantities I will introduce. First is the correlation dimension, d_G and the second is the information dimension d_I . I will simplify these dimension quantities in just suggesting they account for the density of points within the boxes. We can now define a general definition [11] of dimensions of order q :

$$d^{(q)} = \frac{1}{q-1} \lim_{\epsilon \rightarrow 0} \left(\frac{\log_e \sum_{i=1}^n p_i^q}{\log_e \epsilon} \right), \quad (9)$$

where n is the number of phase elements and p_i is the probability that an attractor point falls in the i th element. It is known that d_C , d_I and d_G correspond to $d^{(0)}$, $d^{(1)}$ and $d^{(2)}$ respectively. To learn more about these dimensional quantities, see [12]. As we will find, many fractals will require many measurements to characterise their complex geometry. However, due to limitations of this report, I cannot go into any further detail with these quantities. They represent different methods/approximations in obtaining the dimensions of an object and are therefore, inherently, non-equivalent.

Attractors

I briefly talked about attractors in phase spaces, giving a broad definition of an attractor being the end state for a finite amount of states that describes the system. Here I will go more into detail about attractors, giving a rigorous mathematical definition. We first should start with the idea of *measure*. In one, two and three dimensions, the measure of a solid body E is referred to as the *length*, *area* and *volume* respectively. We are lucky we do not have to deal with any analytical treatment of measure, instead we can leave that to the mathematicians and their investigation of the *problem of measure* [13]. An attracting set, as defined by Milnor [14] is a set A , $A \in \mathbb{R}^n$ satisfying two conditions:

1. The *basin of attraction*, $B(A)$, consisting of all points whose orbits converge to A , has strictly positive measure.
2. For any closed proper subset $A' \subsetneq A$, the set difference $B(A) \setminus B(A')$ also has strictly positive measure.

The first condition states that the basin of attraction must be physically tangible and continuous. The continuous condition comes from the Lebesgue integral definition of measure, where measure can be calculated as the area under the curve for some function. Therefore a set of discrete points will produce zero measure. If we are working in $\mathbb{R}^2$ and $\mathbb{R}^3$ then our basin of attraction will be defined as an area and volume respectively. The second condition refers to how every point in A will be crucial in creating the set. For an attracting set to be considered an attractor, it must satisfy one more condition,

1. it must contain a dense orbit.

Dense orbits are orbits that either passes through, or gets infinitely close with, every point in the attracting set. This condition is imposed to avoid describing an attractor as the combination of two attracting sets.

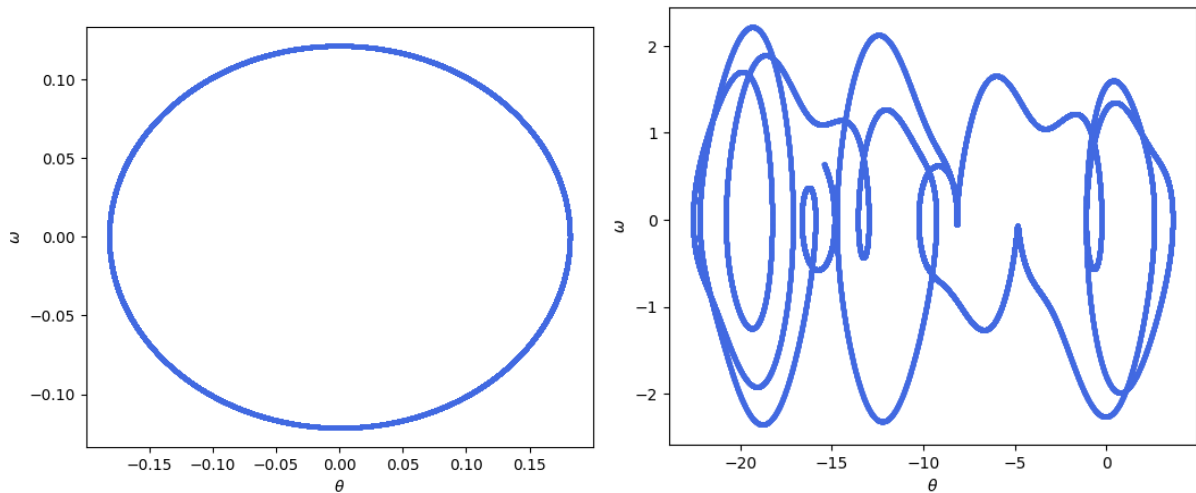
There are three nonfractal/nonstrange types of attractors: *fixed point*, *limit cycle* and *torus* attractors. The torus attractor is beyond the scope of this report and chaotic dynamics in general.

The term *strange attractor* arises from attractors that exhibit chaotic behaviour, that is, sensitivity to initial conditions. However not all strange attractors are chaotic. The idea of *strangeness* comes from the geometric structure of the motion. Strange non-chaotic attractors display fractal geometry in their phase space but lack the extreme sensitivity to initial conditions, therefore is not classified as chaotic. We will be focusing on chaotic, strange attractors in this report.

Fractals

I have constantly referred to this strange, non-Euclidean geometry as fractals but I am yet to define it. We know that our Lyapunov exponents describe the average rate of divergence in a certain direction. Aforementioned, negative exponents correspond to convergence, nonchaotic behaviour whereas positive exponents describe the opposite. Here I will define much more simply, Lyapunov exponents describe the 'stretching' or expansion/contraction in different dimensions. For negative exponents, they describe contraction of the orbits whereas positive exponents represents expansion of the orbits. Exponents equal to 0 describe zero contraction or expansion.

Kaplan and Yorke [15] developed a heuristic, mathematically imprecise definition for a *Lyapunov dimension* [16],



(a) Phase space of a simulated pendulum's motion with parameters, $q = 0.7$, $g = 0.2$ and $\omega = 2/3$

(b) Phase space of a simulated pendulum's motion with parameters, $q = 2$, $g = 1.2$ and $\omega = 2/3$

FIG. 1: (a) An ordered trajectory displaying a stable orbit. (b) Disordered trajectory showing no clear shape. Unpredictable but deterministic.

$$d_L = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|} \quad (10)$$

where j is the largest integer for which $\sum_{i=1}^j \lambda_i > 0$. The condition for strangeness is then reduced to simply d_L must be a non-integer [17]. It is important to note that for non-area preserving/dissipative systems in two dimensions, chaos cannot arise as a result of the Poincare-Bendixon theorem. Therefore we need at least three Lyapunov exponents to observe chaos [18].

Now we can try to understand fractal geometry. A nice analogy would be to think of a crumpled piece of paper in a box. It is a 2D object in a 3D space. The crumpled paper will try to expand (and contract in certain directions) to try to fill the empty space. Of course, our analogy exists in Euclidean space whereas the topic at hand is inherently non-Euclidean but hopefully the imagery helps with the understanding of how fractal geometry comes about.

The term *fractals* comes from *fractional dimension*, coined by Benoit Mandelbrot in 1975 [19]. In Euclidean dimensions, sets can describe ordinary geometric shapes. In 0 dimensional space, sets describe points, for 1 dimensional space, sets describe lines (having only length), in 2 dimensions, sets describe surfaces (length and width), and 3 dimensions, sets describe volumes (having length, width and height). In fractal dimensions, the dimension index can take non-integer values (as seen with the different ways to describe dimensionality), indicating that the set will fill the space. For example, a set with fractal dimension 1.1 will behave like an ordinary line but display some two dimensional thickness. The thickness manifests itself as a self-similarity pattern as you zoom into the 'line' more.

The Chaotic Pendulum

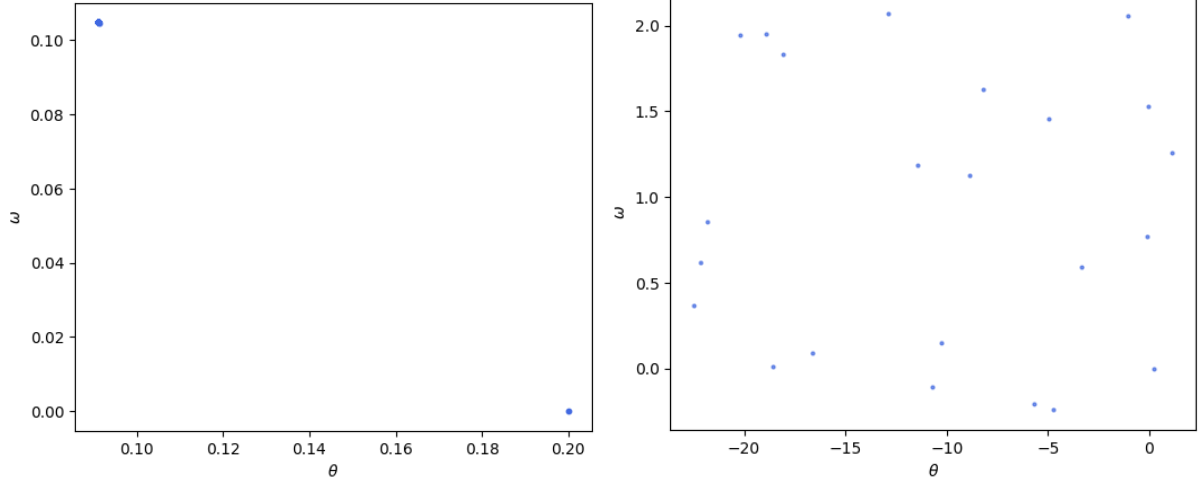
I will utilise a simple case study of the driven pendulum to demonstrate the theoretical concepts aforementioned in the previous sections. The equation for the driven pendulum is given by the third order differential equation [10],

$$\frac{d\theta^2}{dt^2} + \frac{1}{q} \frac{d\theta}{dt} + \sin \theta = g \cos(\omega_D t), \quad (11)$$

where q , g and ω_D are the coefficients of the state variables of the system and different choice of these parameters will produce different trajectories of the system. The state variables are then θ , $\omega = \dot{\theta}$ and ϕ where they represent the position of the pendulum, its velocity and its phase respectively. The equation of motion then clearly satisfies the conditions for chaotic motion, having 3 state variables and a non-linear term being the trigonometric function in the driving force.

I can demonstrate the usefulness of phase space diagrams in characterising non-chaotic versus chaotic by simply tuning the system's parameters as shown in the Figure 1.

We can easily observe a drastic difference in the shape of trajectory of motions between two different sets of state parameters. The stability of orbits can be visually defined as a circular path with no beginning or end whereas the chaotic, unstable orbits do have a beginning and an end and no clear shape or direction in its trajectory. The path is still circular and loopy but the overall shape is undefined, an intrinsic property of chaotic motion. Visually here we can distinguish between periodicity and aperiodicity however in more complicated orbits it may be more beneficial to simplify our phase space. To do this, we will need to define Poincare sections which are essentially



(a) Poincare section of a simulated pendulum's motion with parameters, $q = 0.7$, $g = 0.2$ and $\omega = 2/3$ (b) Poincare section of a simulated pendulum's motion with parameters, $q = 2$, $g = 1.2$ and $\omega = 2/3$

FIG. 2: (a) Poincare section shows two points only located diagonally away from each other on the plot. (b) Poincare section showing multiple points, scattered randomly along the plot.

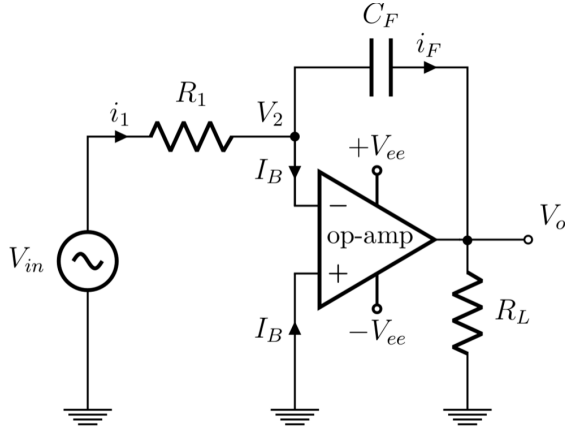


FIG. 3: Diagram showing the layout of an integrator.

The op amp at the centre of the circuit has two inputs, an inverting, denoted by the - symbol and an non-inverting, denoted by the + symbol. There is a capacitor from the output of the op amp that feeds back into the inverting input. The non-inverting input is connected to ground. The input voltage connects to the inverting input via a resistor.

checkpoints for our trajectories at specific locations. Because of pendulum is driven, we can conveniently choose the strobing frequency to match the driven oscillations. This will produce the diagram shown in Figure 2.

The Poincare sections essentially reveal what the phase spaces do but in a more simplistic fashion. We can observe periodicity in the left Poincare plot as when we strobe the states at the same characteristic frequency of the oscillation, the points remain either of two states. This is contrasted in the right plot where when we do the same thing, we find that there are many, maybe even infinite, states possible for our

strobing, a clear sign for aperiodicity.

AIM

The goal of this report is due verify the chaotic dynamics found in a chaotic circuit as proposed by Kiers [20]. I will be using all the analytical tools mentioned above as well as numerical methods that will soon be mentioned to validate the equation of motion for a circuit as a good model for real world observations.

EXPERIMENTAL SETUP

To create a circuit that satisfies the condition for chaotic motion, we need to first understand how to produce an integrated current.

Integrators

By no means am I an electric engineer and so for the scope of this report we will remain in the small realm of physics. Operational amplifiers (Op Amps) are very sophisticated pieces of machinery which in itself deserve their own report but for the sake of explaining how we produce an integrated current, all we need to know is that, for an ideal op amp, the inverting input is kept at the same voltage as the non-inverting input as a virtual ground. In other words, if we take a look at the diagram below 3, we will see a capacitor, C_F , that connects from the output of the op amp to the inverting input.

The purpose of this capacitor is to provide a path for the feedback mechanism of the op amp. The op amp will try to keep the voltage difference between its inverting and non-inverting input equal to 0, that

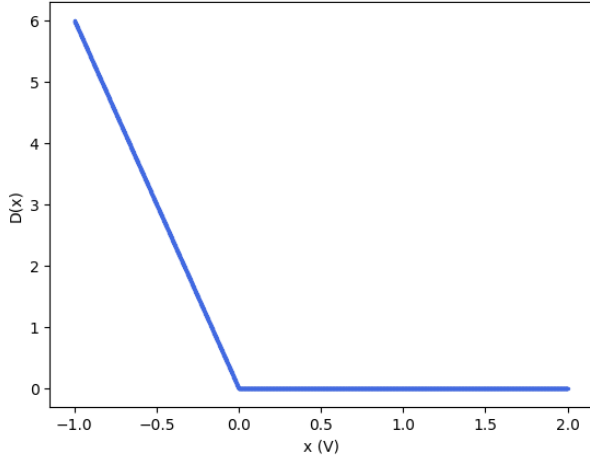


FIG. 4: Theoretical graph displaying the relationship between $D(x)$ and x . The resistors R_2 and R_1 , in line with the underlying paper, are chosen to be in a ratio equal to 6.

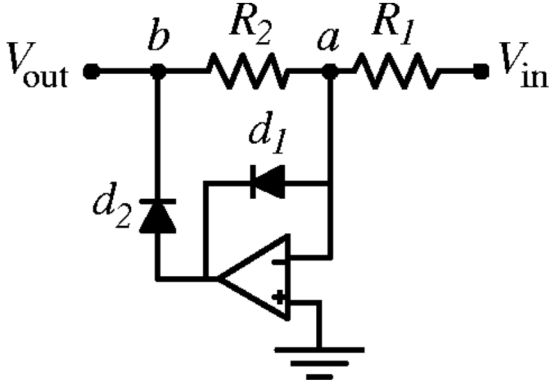


FIG. 5: The non-linear subcircuit is comprised of two diodes and two resistors. Using Shockley's equations [10], one can obtain the relationship between the input voltage and the output voltage to be $V_{out} \approx -\frac{R_2}{R_1} \min(V_{in}, 0)$.

is, $V_- \approx V_+ \approx 0$ and it will do this by adjusting the output of the op amp to produce a current that cancels out the current at the inverting input of the op amp so that there is, in this case, $V_+ = 0$, there is no voltage at the inverting input of the op amp. Using Kirchhoff's law of currents, we have at the inverting input node,

$$i_1 = I_B + i_F. \quad (12)$$

For an ideal op amp, there is no current at the inputs at all so $I_B = 0$ so $i_1 = i_F$. At the capacitor, using the voltage capacitor relationship we have,

$$i_F = C_F \frac{d(V_2 - V_o)}{dt}. \quad (13)$$

and then we can equate

$$\frac{V_{in} - V_2}{R_1} = C_F \frac{d(V_2 - V_o)}{dt}. \quad (14)$$

In an ideal op amp we also have zero voltage at the inverting node so $V_2 = 0$ so

$$\frac{V_{in}}{R_1} = -C_F \frac{dV_o}{dt}. \quad (15)$$

Assuming the output voltage begins at zero, we can rearrange for V_o by integrating both sides to achieve the integrating relationship,

$$V_o = -\frac{1}{R_1 C_F} \int_0^t V_{in} dt. \quad (16)$$

Non-linear component

Now that we have a way to produce integrated state variables, all we need now to produce our chaotic circuit is to introduce a non-linear component. There are a multitude of different non-linear circuits that can be constructed and that are well understood. I will choose to go with the non-linear subcircuit in the reference paper we are verifying which is shown in Figure 5

The output voltage, $D(x)$, can be plotted against the input voltage, x , to show this non-linearity, shown in Figure 4.

The Circuit

We need to satisfy the three state variables condition so we need to have 3 integrators and then a non-linear component. We also need to spend an op amp to couple the integrators together and so therefore, we can construct the circuit as shown in Figure 6.

We need to apply Kirchhoff rules to obtain our differential equation of motion. If we take the output of pin 7 in the 2nd op amp of Chip 2 as x , then the 1st op amp of that same chip will be $-\dot{x}$ and the output of pin 7 in the 2nd op amp of Chip 1 will be $\ddot{x}$. Applying Kirchhoff's law of currents at the summing op amp output we obtain

$$\frac{d\ddot{x}}{dt} + \left(\frac{R}{R_v}\right) V_2 + \left(\frac{R}{R_0}\right) V_0 = -\dot{x} + D(x) \quad (17)$$

or more succinctly,

$$\ddot{x} = -\left(\frac{R}{R_v}\right) V_2 - \left(\frac{R}{R_0}\right) V_0 - \dot{x} + D(x). \quad (18)$$

Here I have omitted the $R_1 C_F$ factor we find in the integrator equation by absorbing it into the time derivative, so we are deriving with respect to dimensionless time, $\tilde{t} = \frac{1}{R_1 C_F} t$.

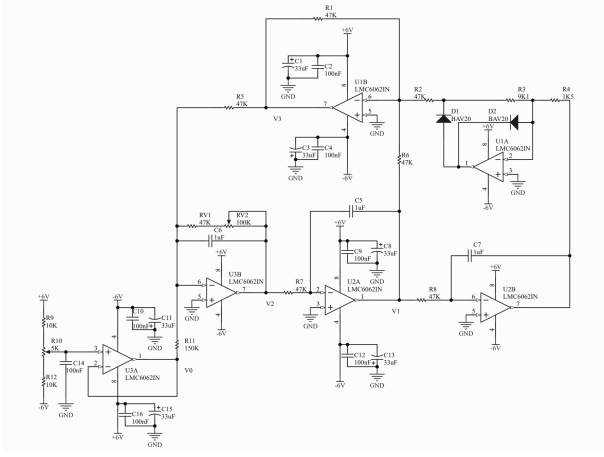


FIG. 6: Circuit diagram of the chaotic circuit. There are 6 op amps in total however LMC6062IN microchips were used. Each chip contained two op amps and the input and output of each op amp corresponding to specific pins on the chip. There are a total of 8 pins on the chip. Pin 3 and 5 corresponded to the non-inverting input of each op amp whereas pins 2 and 6 corresponded to the inverting input of each. Pins 1 and 7 were the output of the two op amps respectively. The three integrators are then the op amps with a $1\mu F$ capacitor placed in parallel, which are the 2nd op amp of the Chip 1 and both op amps in Chip 2. The final op amp, or the 2nd one in Chip 3 serves as a summing op amp.

Numerical Methods (Extension)

The differential equation 18 was solved using a 4-stage Runge Kutta (RK4) algorithm. It is a one step, explicit, numerical integration method with a run time growth of $O(h^4)$. This serves to be a limiting factor in our analysis of chaos as we must choose an appropriate time step for the algorithm that will produce highly resolved trajectory behaviour but also be reasonable in run time. I chose a time step of 0.125 ms for each state solution.

Furthermore, I employed my own version of Wolf's algorithm [21] to calculate all three Lyapunov exponents. Even in this modern day, there is still no stable or accurate way to determine zero or negative Lyapunov exponents in time series data. Wolf's algorithm only allows you to obtain all three if you have the independent equations of each state. This is because in the algorithm we need to know how each state vector evolves with time. We exploit the useful property of Gram-Schmidt reorthonormalisation of preserving orientation of the principal axis whilst also providing the subsequent new orientation of the next state vectors, ensuring that the new axes are orthogonal (no orientation collapse).

Because we know the equations of motion, we can build the Jacobian for the system and observe how it acts on our principal axis. Since we have three state

variables, we have to consider three principal axes vectors,

$$P(t) = \begin{bmatrix} dx(n) \\ dy(n) \\ dz(n) \end{bmatrix}. \quad (19)$$

At each iterative step, the next principal axis orientation will be

$$\begin{bmatrix} dx(n+1) \\ dy(n+1) \\ dz(n+1) \end{bmatrix} = \mathbf{J}_n \begin{bmatrix} dx(n) \\ dy(n) \\ dz(n) \end{bmatrix}. \quad (20)$$

The Jacobian here contains all the divergence information from previous Jacobians, that is $\mathbf{J}_n = \mathbf{J}_{n-1}\mathbf{J}_{n-2}\dots\mathbf{J}_1$. So in each step of the RK4 method, the principal axis vectors are updated with the perturbation from the Jacobian and every couple of steps or so, the axis vectors are re-orthonormalised so that orientation collapse is avoided. The stretching factors in each re-orthonormalisation step in each direction are recorded and the time average is taken to obtain the Lyapunov exponent in that direction.

I used QR decomposition, which is mathematically equivalent to GSR, to obtain the stretching factors in each direction. Essentially, the new perturbed vector,

$$Y_{n+1} = J_n Y_n. \quad (21)$$

The new vector can then be decomposed into

$$Y_{n+1} = QR. \quad (22)$$

Now we treat the new vector as the new basis components, Q ,

$$Y_{n+1} \rightarrow Q \quad (23)$$

and the process is restarted. Here, I only re-orthonormalised, that is, setting the vector as Q , once every 10 steps. The R matrix is an upper triangular matrix that contains all the stretching factors in all directions. The growth rates are found in the diagonal components of the matrix and so the average can be found by the taking the summation over all R matrices in each step.

Even to this day, Lyapunov exponents that are 0 or negative are impossible to calculate in an accurate or stable manner. Therefore, we can only calculate non-negative exponents, which will be done using the *nolds* Python package, which implements the Rosenstein algorithm [22].

RESULTS AND DISCUSSION

I will admit that it was my first time constructing a more involved circuit. I had, up until this point, never

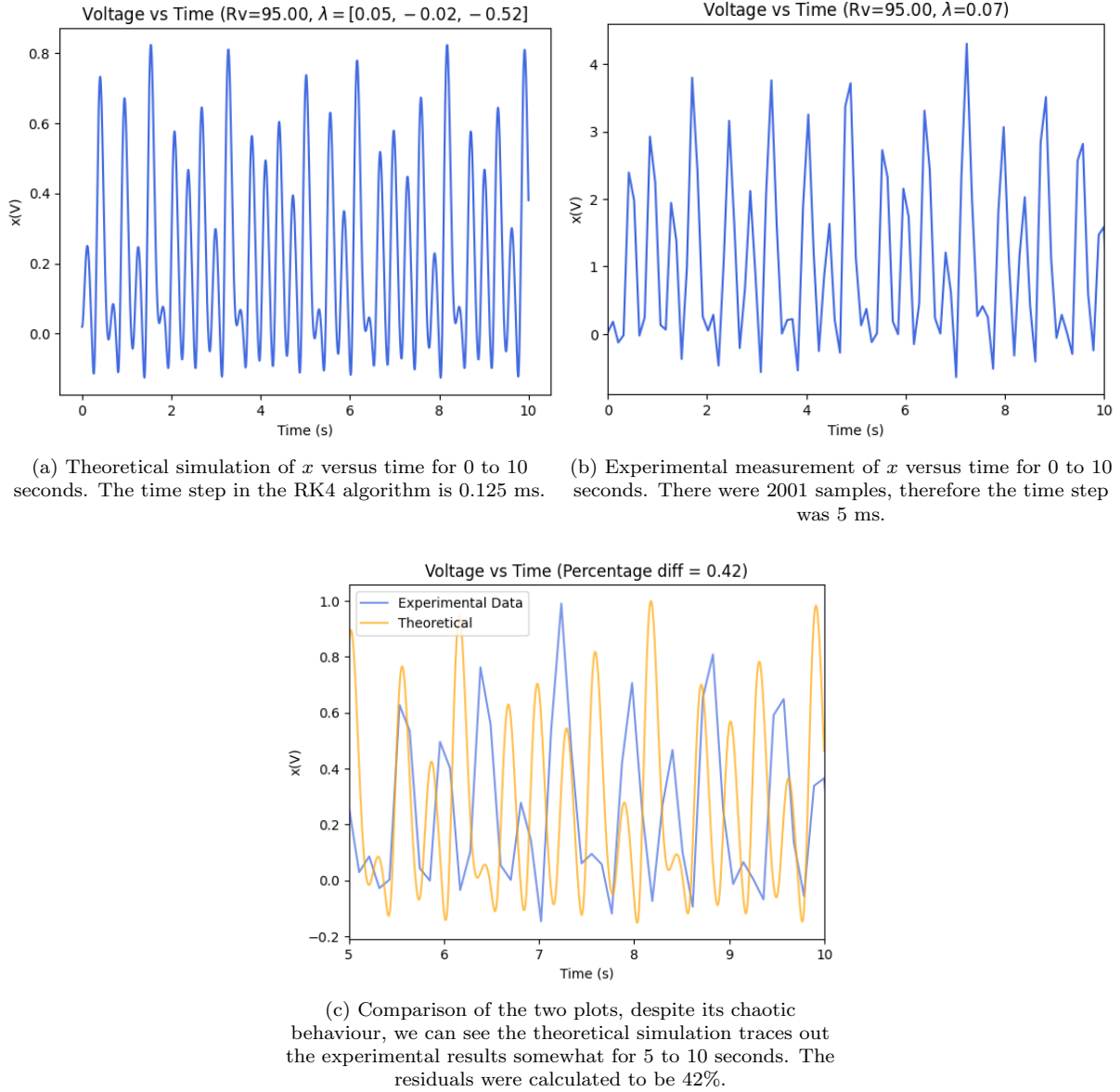
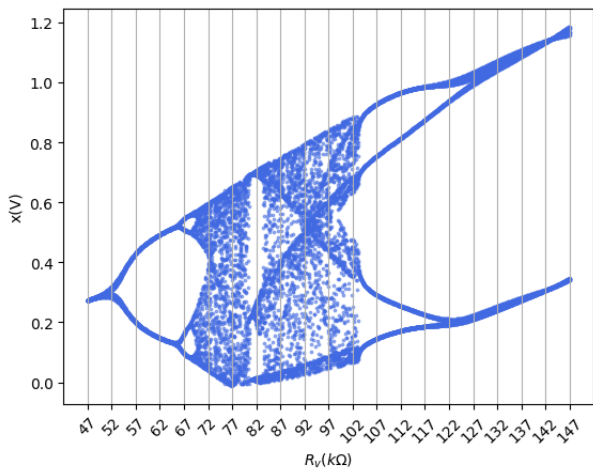


FIG. 7: We want to see how well the simulation matches experimental results. The simulation shows higher characteristic frequency of oscillation than experiment. There is clear chaotic dynamics going on because there is one positive, one near zero and one negative Lyapunov exponent. The experimental Lyapunov exponent weakly matches the non-negative theoretical component, approximately a 40% disagreement, in the same range as the error found in calculating residuals.

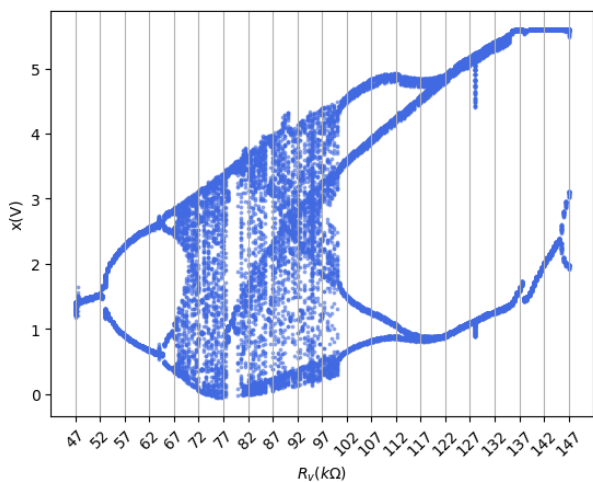
used a breadboard nor connected any wires in any way and therefore, the construction of the circuit in Figure 6 proved to be quite cumbersome. I am grateful to have undertaken this journey as I have learnt plenty and thoroughly enjoyed the process. However, ultimately, I believed there to be some error in the final iteration of the circuit, and the iteration for which the experimental data was taken. This clear error, due to human mistake, can be clearly depicted in Figure 14 in Appendix A. Before the circuit was connected wholly, we tested each Chip individually for range of voltages. I found that, along with many of my other peers, the non-linear subcircuit output produced voltages in the range of -1 to 6 volts, which was not quite the 0 to 6 volts range we were modelling from. The idea of

'good enough' was eventually decided upon, meaning the circuit was not perfectly matching the theoretical simulation but I wanted to show the behaviour was ultimately the same. I knew going ahead meant the quantitative analysis would not be perfect but the circuit should still display chaotic behaviour similar to theoretical expectations. It seemed, at first glance, that the experimental data matched theoretical simulation decently well, shown in Figure 7.

In the overlay plot, we can see that the shape of the curves do match, although the experimental result seemed to be oscillating a bit slower. I can attribute this to the lower sampling frequency of the LabJack settings. If this experiment were to be re-done with more time investment, the data could have



(a) Bifurcation points at 52 kΩ, 65 kΩ, 68 kΩ, 103 kΩ, 125 kΩ.



(b) Bifurcation points, approximately, at 52 kΩ, 64 kΩ, 68 kΩ, 100 kΩ, 113 kΩ and 118 kΩ.

FIG. 8: The top plot is the theoretical simulation of the bifurcation plot. We find a smooth curve with clear regions of nonchaos and chaos. The bottom plot is the bifurcation plot of experimental results. We find strange artefacts in measured peaks. These weird artefacts tend to occur at bifurcation plots. There is also another bifurcation point at 145 kΩ in the experimental plot, which is a strange artefact rather than reflective of the underlying physics. There also is a saturated region in the top branch of the experimental bifurcation plot at 135 kΩ.

been more finely resolved. The code for the Lab-Jack was completed by myself and looking retroactively, could have been much improved in efficiency. A rough estimation of disagreement between the theoretical and experimental results can be calculated from the residuals, being approximately, a 42% deviation. This lines up with the Lyapunov exponent calculations, being approximately 40% deviation. The calculation of the non-negative Lyapunov exponent from time series data is a complicated process, involving advanced statistical methods such as Monte Carlo sampling and therefore the process is beyond

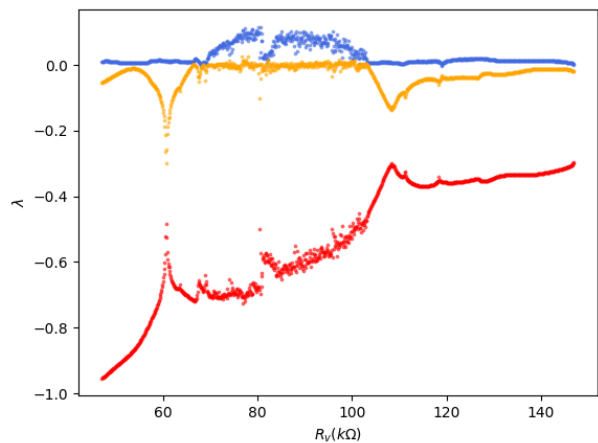


FIG. 9: Calculated Lyapunov using the Wolf algorithm for theoretically simulated results. We see in regions where there are 2 negative exponents and 1 zero exponent, we have a limit cycle attractor. In regions of chaos, it can be defined as the regions where there are 1 positive, 1 zero and 1 negative exponents.

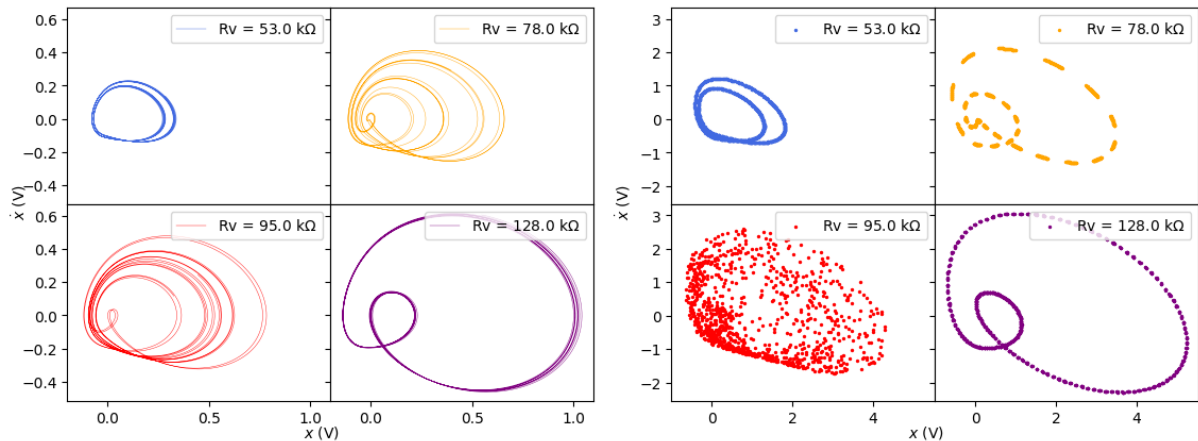
the scope of this report. Therefore, I cannot give a proper uncertainty on its calculation. For the calculation of theoretical Lyapunov exponents, I can more fairly estimate the uncertainty as the process is more known. I will use block averaging [23] to find the average standard deviation of the Lyapunov dataset. For the chaotic attractor, $Rv = 95$ kΩ, the largest Lyapunov exponent is calculated to be $\lambda_1 = 0.059 \pm 0.004$, the second largest, $\lambda_2 = 0.0027 \pm 0.003$ and the smallest, $\lambda_3 = -0.554 \pm 0.003$. Therefore, if we apply the Kaplan-Yorke equation, we obtain a value for the dimensionality of the circuit,

$$d_L = 2 + \frac{0.059 + 0.0027}{0.554} = 2.111 \pm 0.124. \quad (24)$$

Note here that the large uncertainty originates from the second Lyapunov exponent being close to zero. Nevertheless, we observe behaviour just a bit around 2 dimensions, an exclusive indicator for fractal behaviour which is expected.

The chaotic indicator of 1 positive, 1 zero and 1 negative exponent can clearly be seen in the theoretical plot seen in Figure 9. We find that in regions of nonchaos, the exponents behave in quite stable manner but once we get into that chaotic region, roughly above $Rv = 65$ kΩ and below $Rv = 105$ kΩ, the Lyapunov exponents start to become chaotic themselves. All three exponents, start to fluctuate in seemingly disorderly. From previous work, we understand that the larger the positive exponent, the more chaotic behaviour is observed. For example, in the small region where the largest exponent is approximately zeroed in between the chaotic regions, in the region of $80 < Rv < 85$, we can observe slightly nonchaos. This is better shown in Figure 8.

The theoretical bifurcation plot agrees quite



(a) Theoretically simulated plots for variable resistance set to 53, 78, 95 and 128 kΩ. A clear limit cycle attractor is shown in the top left and bottom right plots, indicating regions of nonchaos. The orbit appears to cross however this is a 2D cross section of a 3D plot and so the loop never intersects with itself in space. The chaotic strange attractors are shown in the top right and bottom left plot. We find there are trajectories that come really close to each other, however they never touch. In fact, they get infinitely close to each other but will never touch, this is fractal behaviour.

(b) Rather than plotting experimental data as lines, we choose to create a scatter plot of trajectories. This is because, our time step, experimentally, was too high to accurately resolve trajectories. Creating a scatter plot allows us to show the generic shape of the attractor. We can see all panels except the bottom left plot exhibit stable orbital behaviour. The bottom left plot clearly shows chaotic behaviour with no clear circular trajectory or shape.

FIG. 10: We find that all plots except the plot for $Rv = 78$ kΩ agree between experimental and theoretical simulated data. The experimental result for the chaotic attractor, $Rv = 95$ kΩ displays more chaos than theoretically expected. We can tell as the states fills out most of the circular shape of the attractor whereas in the simulated result, there is some empty space within the circular attractor.

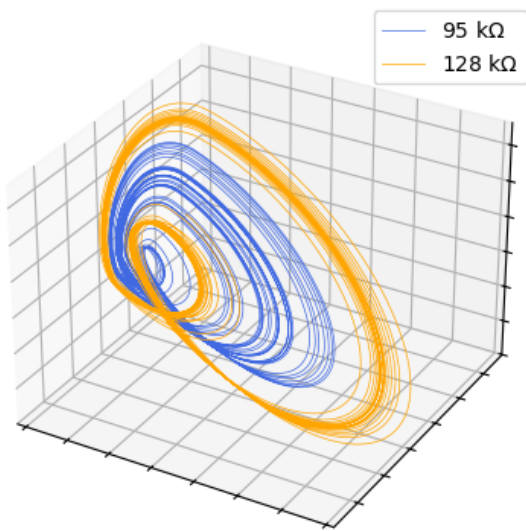
strongly with the Lyapunov exponent graph. We begin with a stable plot from $Rv = 47$ kΩ to $Rv = 53$ kΩ. We then see period doubling until approximately $Rv = 68$ kΩ and then chaos. In regions of 1 zero and 2 negative Lyapunov exponents, limit cycle attractors will be observed. These are stable orbits are however then doubled in period, reflected in the two negative exponents slightly converging before entering the state of chaos. Past the chaotic regime, we will observe period halving and then stable orbits again, still with limit cycle attractors. We find this behaviour to still be reflected in the bifurcation plot produced with experimental data, although slightly shifted towards the right. The overall behaviour still remains the same but there are oddities, especially at bifurcation points and then near the end of the graph, where the peaks seem to be saturated, capped at 6V. The bifurcation points do somewhat agree, and to be honest, better than expected due to the differences found in previous plots. A further investigation into specific Rv trajectories are found in Figure 10. The values for Rv were chosen based on one nonchaotic, one chaotic, one where the theoretical and experimental disagree and then one post the chaotic regime.

The phase space plots better reflect and visualise the attractors I have been talking about. The strange chaotic attractors are much more obvious to see and distinguish from the limit cycles. Even in the experimental data, we can observe that limit cycles greatly restrain the region in which states can exist, creating high density trajectories that produce a circular

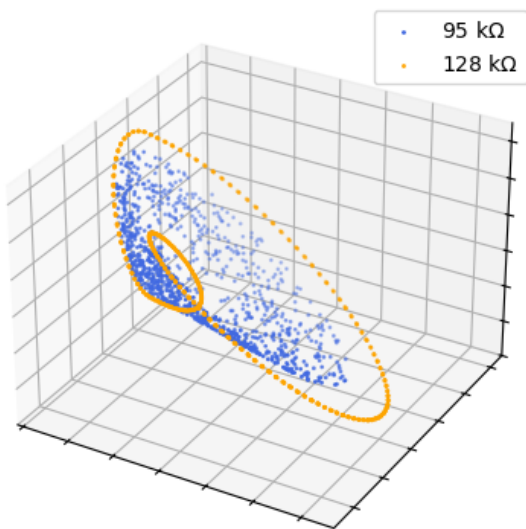
or loop shape. We can observe that even though in the 2D plots, the loop seems to cross over itself, this does not violate the rule put forth by the phase space conventions. We can better see this non-violation in Figure 11.

The loop possesses a 3D depth that is unable to be expressed in a 2D plot but in 3D we can better see this characteristic. The plot of both attractors deceptively show the violation where trajectories cross over each other but rather this is exactly what fractal behaviour tends to do. In reality, these trajectories get infinitely close together and weave themselves through one another at an infinitely small scale whereby the pattern is self similar and therefore the trajectories can pass 'through' each other without touching. We can sort of see this behaviour in a Poincare section where rather than stroboscopically probing the plot at the same frequency or a driven frequency, we observe when the plot crosses a certain plane, seen in Figure 12.

The Poincare plots reveal that chaotic trajectories 'revisit' the same plane but at a slightly different angle/state. This infinitely small change in state is then the fractal behaviour. Finally I want to end the investigation of this behaviour by displaying another obvious sign of chaos. Despite the multitude of noise that can arise from circuit behaviour, plotting the oscillations in Fourier space reveals that in non-chaotic regimes, the signal is picked up quite easily and strongly. Therefore, the 'noise' seen in the third plot in 13, is actually induced wholly by chaotic be-



(a) Simulated result



(b) Experimental data

FIG. 11: A 3D plot showing the chaotic attractors $Rv = 95 \text{ k}\Omega$ and $Rv = 128$ for both the simulated and experimental results. The constant looping around has depth in the $\ddot{x}$ axis. We can more clearly see how each trajectory do not cross over itself however showing this plot to the uninitiated, the condition that trajectories cannot cross each other would seem to be broken.

haviour, an intrinsic characteristic of the dynamics, not influenced by external factors.

This kind of insight shows that in chaotic systems, the 'noise' that can be observed is largely generated endogenously rather than an amplification of generic noise.

The return maps for the simulated result are shown in Figure 15 however experimental curves could not be produced due to insufficient duration of sampling. Overall, the theme of this investigation did not require

the fractal behaviour exhibited by these return maps, I find that fractal behaviour is better reflected in the plots shown above.

CONCLUSION

Despite the trials and tribulations in the journey of this report, I still managed to strongly show chaotic behaviour in the circuit built. Although the circuit did not quantitatively reflect the simulated model, plots showing chaotic behaviour could be produced and the notion that these regions of chaos existed experimentally were convincingly conveyed. Therefore, the aim of this experiment to validate the findings of Ref [10] were satisfied and to an acceptable extent. Basic plots for chaotic systems such as phase spaces and Poincare sections could be produced for the experimental setup and advanced techniques in analysing chaotic behaviour could be conducted such as producing bifurcation plots and calculating Lyapunov exponents. I found that the chaotic circuit proposed by Ref [10] exhibited near to, but not exactly, 2 dimensional behaviour which agrees with their result and conclusion.

ACKNOWLEDGEMENTS

I would like to forward my many thanks to Nicholas for helping me build the circuit during the lab and would like to shout out his endless patience. I would also like to give my many thanks to my lovely girlfriend whom have been a great supporter in me writing this report.

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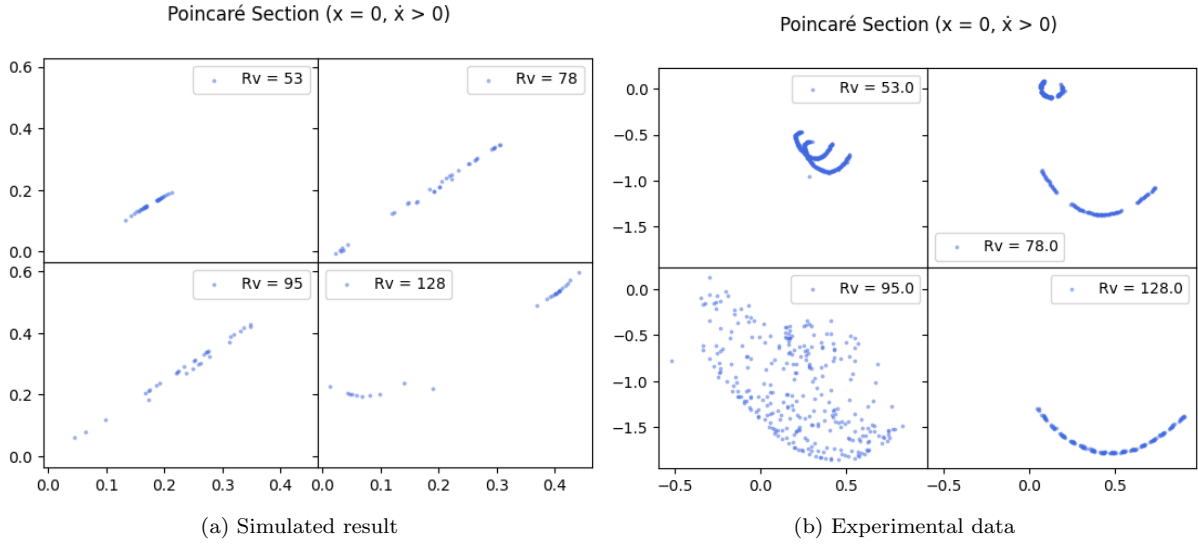


FIG. 12: Poincaré sections at $x = 0$, coming from positive $\dot{x}$. The shape of the plane is different between simulated and experimental results. Theory predicts that the plane of intersection is a straight diagonal line, however the experimental data demonstrates that the plane is more of a concave up arc. The behaviour of Poincaré sections for each Rv are the same across theory and experiment. We find that in regions of nonchaos, there are clear defined regions, exhibited by dark blue regions indicating high density of states. As Rv reaches regions of chaos, the density of states decreases and more spread out across the plot. In the theoretical Poincaré section, we find a second layer of the plane, indicating small increased dimension of a 2D plane.

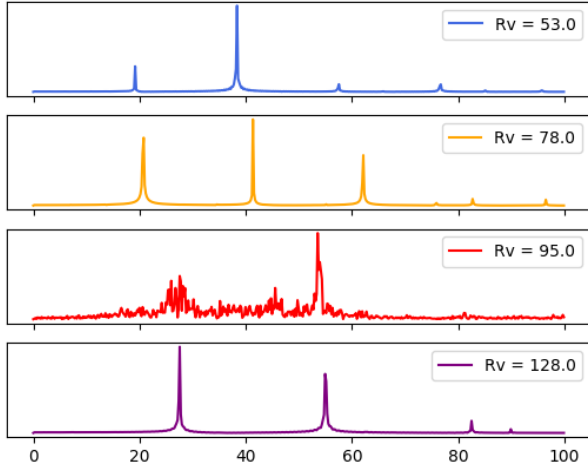


FIG. 13: Fourier space plots for experimental data.

In regions of nonchaos, there are narrow band signals, clearly indicating periodicity. For the spectrum in $Rv = 95$, there is supposedly noise, however we know it is not noise as we do not see it in the other plots.

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Appendix

Non-linear subcircuit

Return Maps

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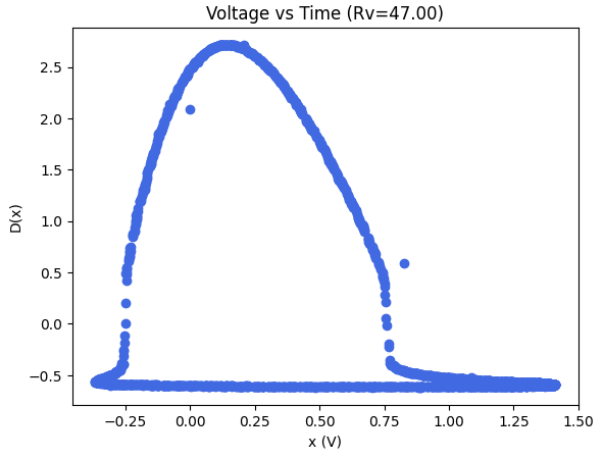
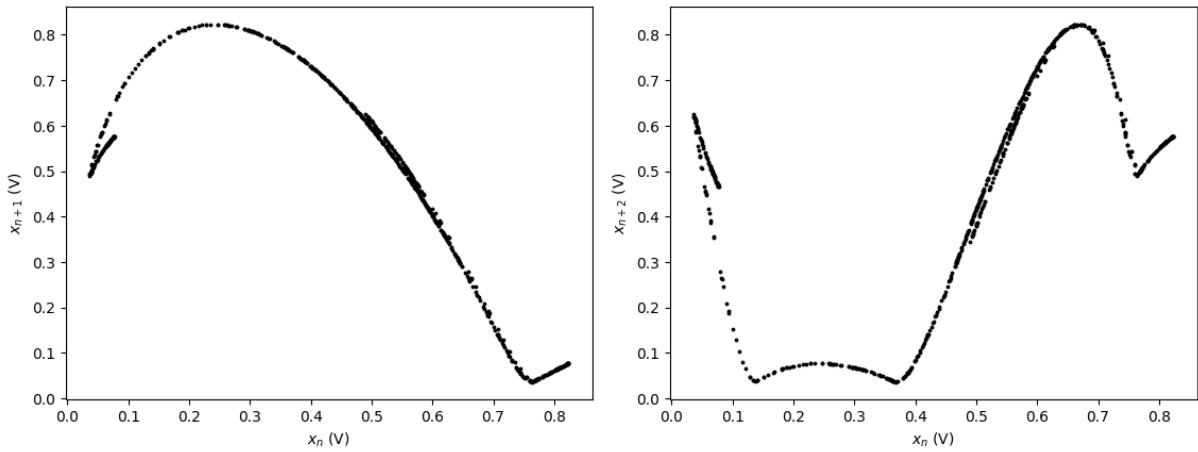


FIG. 14: Graph displaying experimental measurement of the non-linear component. The lack of drop off shows weak diode connection. However there is a non-linear feature as the subcircuit does not allow voltages past 0.75V, instead outputs -0.5V. The overall shape is not expected and therefore the simulated results and this experimental data will not be valid.



(a) Return map for simulated result for $n+1$ th peak for n th peak. (b) Return map for simulated result for $n+2$ th peak for n th peak.

FIG. 15: To show a complete return map with fractal structure, we need 20 times more sample points than previous RK4 solutions. This is to generate more peaks and so when we were generating experimental data, we could not produce a return map due to scarcity of peaks.